

ARAB YOUTH SPACE HACKATHON 2026 · CHALLENGE 813 · THEME 02

QAYDH

Urban heat risk, through the eyes of satellites

Theme 02 · Urban Expansion, Land Use Change & Heat Risk

Team T0049 · United Arab Emirates

TEAM T0049

Three people, one city problem

Entesar Alhebsi

Team lead · story, README, submission

Noora Alhajeri

Notebook, data, models, dashboard

Maryam Albonni

Deck, licences, checks

THE PROBLEM WE SOLVE

Heat maps exist. Decisions don't.

What exists today

- City heat maps: they show where, not who or why
- Starter NDBI rule: calls 92% of Musaffah "built"
- Site surveys: slow, one district at a time
- Generic advice: "plant trees" everywhere

What QAYDH adds

- People: who is outside, at which hours
- Surfaces: road, roof, sand, green at 10 m (F1 0.91 on pure pixels)
- Every Gulf city, every summer, open data
- One named action per hotspot, with its reason

MUSAFFAH, ABU DHABI

Real places, not pixels



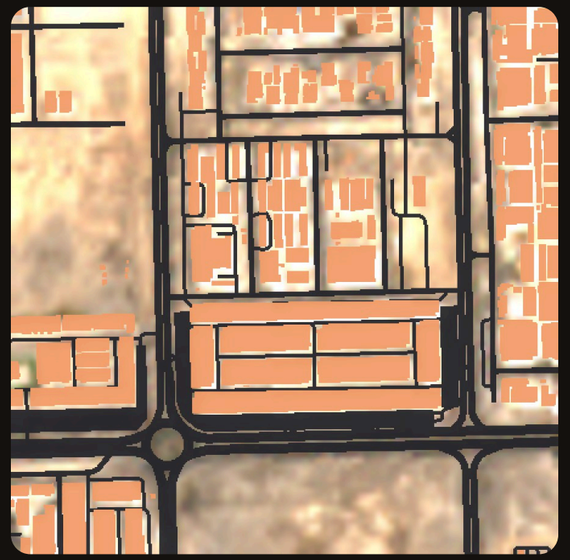
Hotspot M-001

P93 heat · 54% pavement



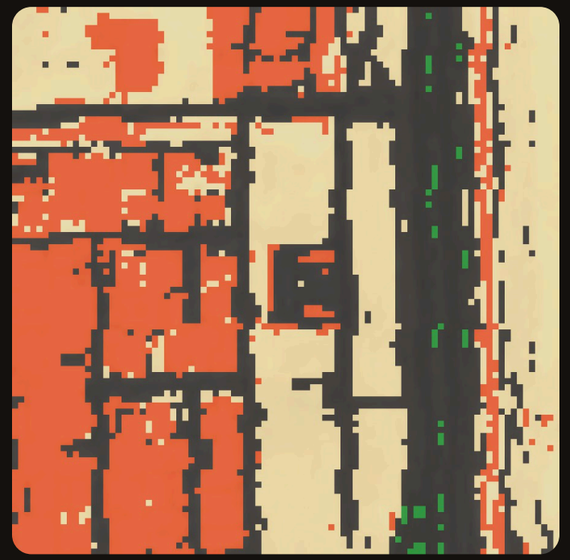
Bus stop · Bus stop on Mohammed Bin Kha

P97 heat · 0.0% green



Mosque · Mosque inside camp

P75 heat · nearest green 306 m



What is physically there

road · roof · sand · green at 10 m

Sentinel-2 10 m · buildings: Microsoft + OSM · roads: OSM · heat: Landsat

BUSINESS USE CASE · THE END USER

Who loses when a city heats up, and what do they do today?

Municipal planners

TODAY

One-off consultant studies, complaints, the same tree programme everywhere

WITH QAYDH

A ranked list of streets and sites, each with its reason and its fix

Industrial zones & contractors

TODAY

One blanket midday-break rule; no idea which yard or camp is worst

WITH QAYDH

Which yards, camps and bus stops need shade, rest nodes and roof coating first

Transport & delivery operators

TODAY

Riders and passengers wait wherever the stop or restaurant happens to be

WITH QAYDH

Cooling points and shaded stops placed where heat and waiting overlap

8.3 h
a day above 40 °C air

87/92
summer days at 40 °C+

+15 °C
surface–air difference at 10:40 (descriptive)

157,076
residents in the hottest fifth of east Riyadh (transfer test)

Business model · per-city subscription: dashboard + API + summer report · open-data core keeps cost near zero

WHO BENEFITS

Built for the people who decide where shade goes

Municipalities & planners

Ranked hotspots, matched actions, summer briefs

Transport authorities

Which bus stops to shade and cool first

Developers & new districts

Heat-aware master plans before roofs lock in

Delivery & mobility platforms

Rider cooling points and safe hours

Research centres & universities

Open, reproducible urban-heat evidence

Space & EO programmes

A ready urban use-case for Satellite 813

Logos show example target users; no partnership or endorsement implied.

OPEN DATA WE BUILT ON

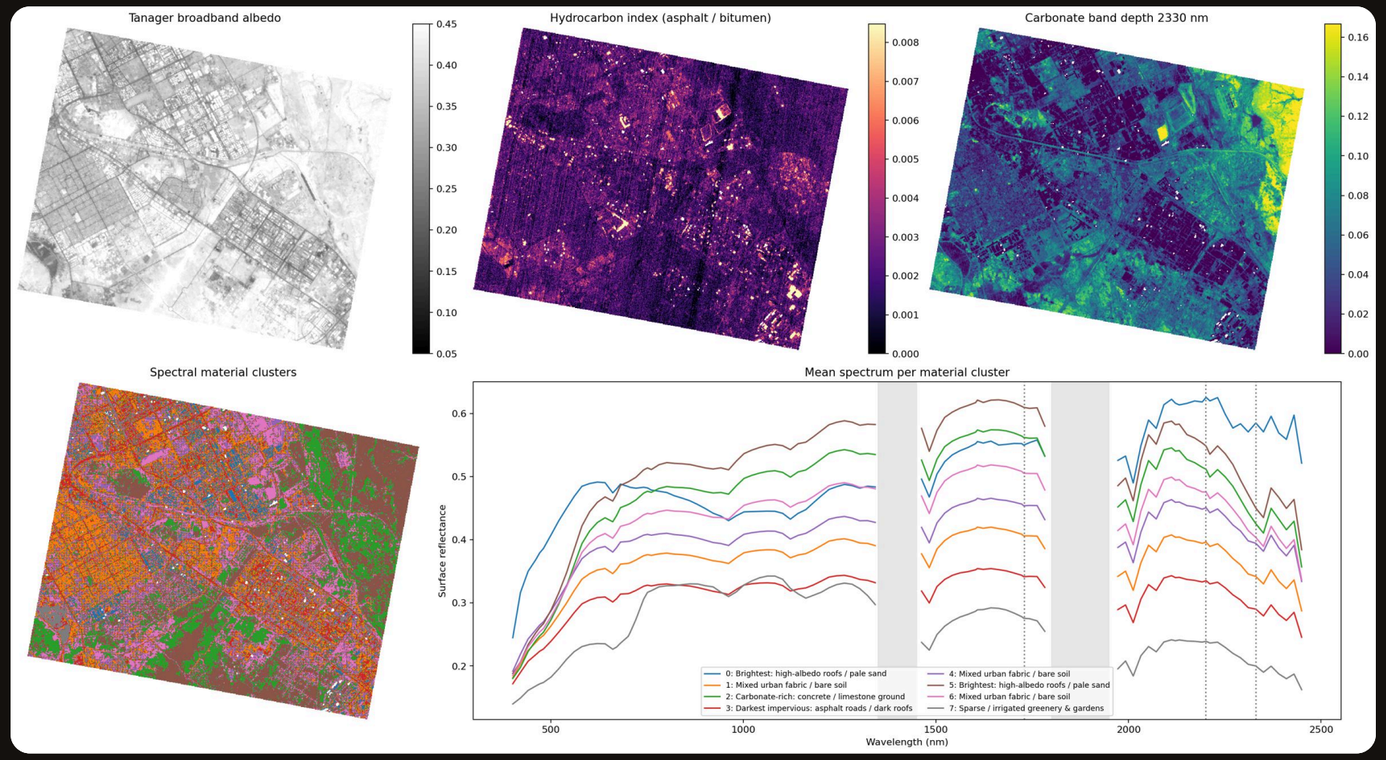
Nine open sources, one answer

USGS science for a changing world	Landsat 8/9 thermal + optical
esa	Sentinel-1 radar · Sentinel-2 10 m · WorldCover
planet.	Tanager hyperspectral (426 bands)
NASA	EMIT hyperspectral over Musaffah
	Roads · bus stops · mosques · schools
Microsoft	25,462 building footprints
WorldPop	Residents per 100 m
ECMWF	ERA5 hourly weather
NOAA	Station ground truth

Logos identify data providers; no endorsement implied.

HYPERSPECTRAL · PLANET TANAGER 426 BANDS

Narrow bands read the material, not just the colour



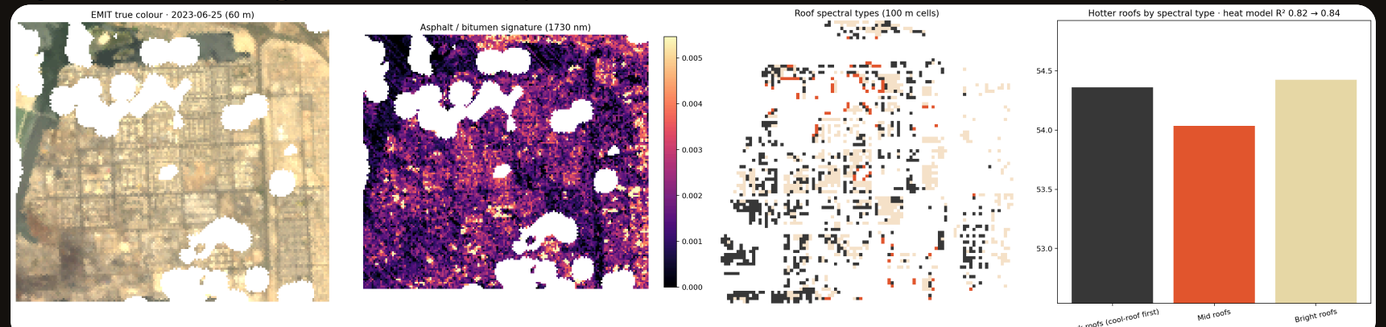
+0.13
R^2 for heat when Tanager is added (built-up, spatial CV)

0.76
surface F1 on unseen tiles (road · roof · sand · green)

r 0.87
Tanager ↔ Landsat albedo after co-registration

Abu Dhabi · EMIT
Musaffah heat model R^2 0.82 → 0.84 with NASA EMIT hyperspectral

Riyadh · Planet Tanager solar albedo · asphalt 1730 nm · concrete 2330 nm · material clusters



ARCHITECTURE

From open satellites to a decision

Data

- Landsat 8/9 thermal
- Sentinel-2 10 m
- Planet Tanager hyperspectral
- WorldCover · Microsoft footprints
- OpenStreetMap · WorldPop
- ERA5 + NOAA stations

Process

- Cloud & quality masks
- Annotation ≥80% purity
- SAM segments · RF models
- Spatial validation A–E

Answers

- Hazard zones
- Surface mix
- Exposure sites
- Heat drivers
- Priority + action

Delivery

- Story dashboard
- GeoJSON / API
- Planner brief per hotspot
- Summer alerts

MATERIAL → CAUSE → FIX

Every material gets its own fix

MATERIAL	HOW SATELLITES SEE IT	WHY IT IS HOT	REPLACE / TREAT WITH	WHERE
Metal sheet roof	flat SWIR, no 2330 nm dip, corrugation texture	Heats fast, re-radiates into rooms and street	White high-SRI coating + under-deck insulation	11% of roofs (rough estimate)
Concrete / cement roof	carbonate dip at 2330 nm, mid albedo	Stores heat, releases it after sunset	Elastomeric cool-roof coating (Estidama SRI ≥ 78)	57% of roofs (rough estimate)
Asphalt road / car park	hydrocarbon dip at 1730 nm, albedo < 0.12	Darkest surface; hot into the night	Street trees + shade first; cool-pavement seal coat (minor)	54% of M-001
Bare sand lot	bright, Fe^{3+} slope 500–900 nm, no green	Radiates heat to the air around it	Shade sails + Ghaf / Sidr on TSE drip	44% of M-001
White / cool roof	albedo > 0.45	Already reflective	Keep: clean and recoat	24% of roofs (rough estimate)
Vegetation	red edge 705–750 nm, NDVI > 0.3	Cools 5.3 °C vs built-up (Riyadh)	Protect and extend into hotspots	the coolant

HOW TO USE THE DASHBOARD

Open it, click, decide

1 · OPEN

noora-alhajeri.github.io/QAYDH-813-T0049 — no login, any browser

2 · PICK A CITY

Tabs at the top: East Riyadh · Abu Dhabi · **Musaffah deep dive**

3 · WALK THE STEPS

Next / Back (or ← →) moves through 6 steps: where · who · what · labels · why · act

4 · CLICK A PLACE

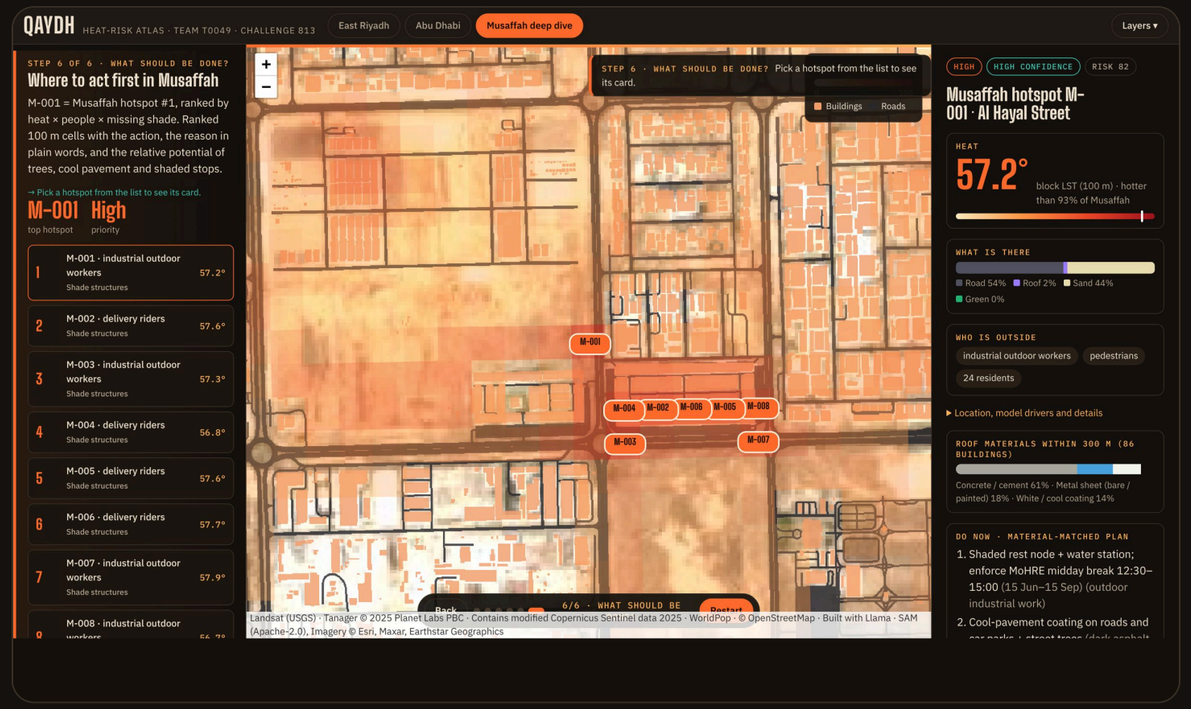
B bus stop, M mosque, S school → card with heat %, greenery and the fix

5 · CLICK A HOTSPOT

M-001, M-002 … → materials, who is outside, do-now plan

6 · LAYERS ▾

Switch heat zones, surfaces, buildings, priority; click a building for its block temperature



STEP 1 OF 6 · WHERE IS HEAT HIGH?

Hazard zones, not fake street temperatures

Landsat thermal (≈100 m) shows zones: elevated ≥ 56.0 °C, high ≥ 57.0 °C, extreme ≥ 57.5 °C. It never pretends to know one bus stop's exact temperature.

→ Orange = elevated · red = high · dark red = extreme (top 5%). Press Next to see who is there.

57.5 °C extreme threshold (P95)

55.0 °C block median

STEP 2 OF 6 · WHO MAY BE EXPOSED?

Bus stops, mosques, clinics, labour camps

STEP 3 OF 6 · WHAT IS PHYSICALLY THERE?

Ro… m

③ Steps 1→6 on the left

STEP 4 OF 6 · HOW WE LABELLED IT

Annotation, objects and AI segments

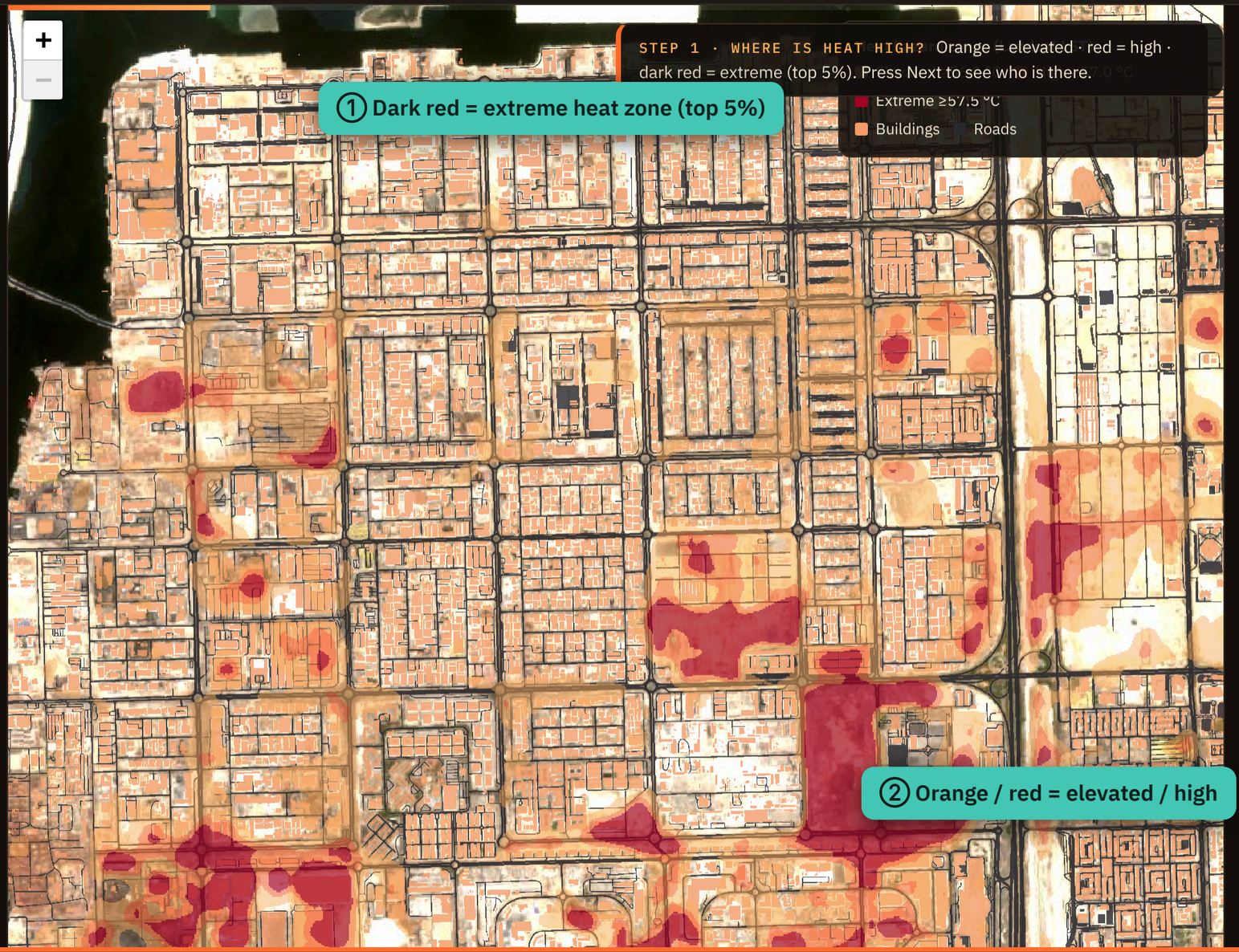
STEP 5 OF 6 · WHY MAY IT BE HOT?

Asphalt and sand up, green down

STEP 6 OF 6 · WHAT SHOULD BE DONE?

Where to act first in Musaffah

1	M-001 · industrial outdoor workers Shade structures	57.2°
2	M-002 · delivery riders Shade structures	57.6°



Musaffah deep dive

Abu Dhabi · 9 × 9 km at 10 m

WHERE	Persistent summer hotspots at 300 m
WHEN	Hours above 40 °C, from weather reanalysis
WHY	Roof, road, sand or missing greenery, read from hyperspectral spectra
WHO	Residents + people outdoors: mosque-goers, bus riders, delivery riders, workers, children
WHAT NEXT	One named action per hotspot, with estimated °C and confidence

Open the last chapter, “Where to act first”, and pick a hotspot.

Open data: Planet Tanager (© 2025 Planet Labs PBC), Landsat 8/9, ESA WorldCover, WorldPop 2025,

WHERE IS HEAT HIGH?

01 Hazard zones, not fake street temperatures

Extreme ≥ 57.5 °C. Landsat gives the zone; buildings and places tell us what falls inside it.

STEP 2 OF 6 · WHO MAY BE EXPOSED?

Bus stops, mosques, clinics, labour camps

139 named outdoor sites from OpenStreetMap, each scored within 150 m: heat percentile, vegetation, impervious cover, distance to green.

→ Click a lettered icon: B bus stop, M mosque, S school/clinic.

1 very-high sites

7 high sites

STEP 3 OF 6 · WHAT IS PHYSICALLY THERE?

Road, roof, sand, plants, water at 10 m

STEP 4 OF 6 · HOW WE LABELLED IT

Annotation, objects and AI segments

STEP 5 OF 6 · WHY MAY IT BE HOT?

Asphalt and sand up, green down

STEP 6 OF 6 · WHAT SHOULD BE DONE?

Where to act first in Musaffah

1	M-001 · industrial outdoor workers Shade structures	57.2°
2	M-002 · delivery riders Shade structures	57.6°
3	M-003 · industrial outdoor workers	57.3°



VERY HIGH | BUS STOP

S-001 · Bus stop · Mohammed

② Card: heat %, green, action

HEAT	P97 within 150 m
GREEN	0% vegetation · nearest green 230 m
SURFACE	64% impervious
CONTEXT	public transport

RECOMMENDED

Shaded, ventilated bus shelter with cooling

Why: people wait unshaded at peak heat.

02

WHO MAY BE EXPOSED?

139 named outdoor sites

Each scored within 150 m: heat percentile, green, impervious cover, distance to shade.

STEP 3 OF 6 · WHAT IS PHYSICALLY THERE?

Road, roof, sand, plants, water at 10 m

Rule-based reference labels from OpenStreetMap, Microsoft footprints and WorldCover, kept only at ≥80% purity, mixed pixels excluded, noisy labels removed, scored on held-out 1 km blocks.

→ Grey road, violet roof, beige sand, green plants, blue

② Accuracy on unseen blocks

0.91 macro-F1, held-out blocks

0.47 starter index rules

STEP 4 OF 6 · HOW WE LABELLED IT

Annotation, objects and AI segments

STEP 5 OF 6 · WHY MAY IT BE HOT?

Asphalt and sand up, green down

STEP 6 OF 6 · WHAT SHOULD BE DONE?

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Open data: Planet Tanager (© 2025 Planet Labs PBC), Landsat 8/9, ESA WorldCover, WorldPop 2025,

03

WHAT IS PHYSICALLY THERE?

Road · roof · sand · green · water

Held-out blocks, pure pixels: macro-F1 0.91 (starter rules 0.47). On random points 0.42: classes are indicative.

STEP 4 OF 6 · HOW WE LABELLED IT

Annotation, objects and AI segments

Reference labels at ≥80% purity (A–C train · D validate · E test), 25,462 building objects with ID and bounding box, SAM segments around hotspots, and a 300-point review queue for people to check.

→ Hover a building outline to see its ID and labels. Orange outline = cool-roof candidate.

25,462
labelled building objects

72
SAM annotation candidates

STEP 5 OF 6 · WHY MAY IT BE HOT?

Asphalt and sand up, green down

STEP 6 OF 6 · WHAT SHOULD BE DONE?

Where to act first in Musaffah

1	M-001 · industrial outdoor workers Shade structures	57.2°
2	M-002 · delivery riders Shade structures	57.6°
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Musaffah deep dive

Abu Dhabi · 9 × 9 km at 10 m

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Open data: Planet Tanager (© 2025 Planet Labs PBC), Landsat 8/9, ESA WorldCover, WorldPop 2025,

03b

HOW WE LABELLED IT

Objects with ID, box and label

25,462 buildings labelled: roof class, heat zone, dark-roof flag, street. 445 are cool-roof candidates.

STEP 5 OF 6 · WHY MAY IT BE HOT?

Asphalt and sand up, green down

A spatially cross-validated model links each 100 m cell's surface mix to its heat. Drivers are shown as associations, not proof of cause.

→ Each 100 m block is coloured by its summer surface temperature (yellow cooler → dark red hotter). Click a building to see its block value.

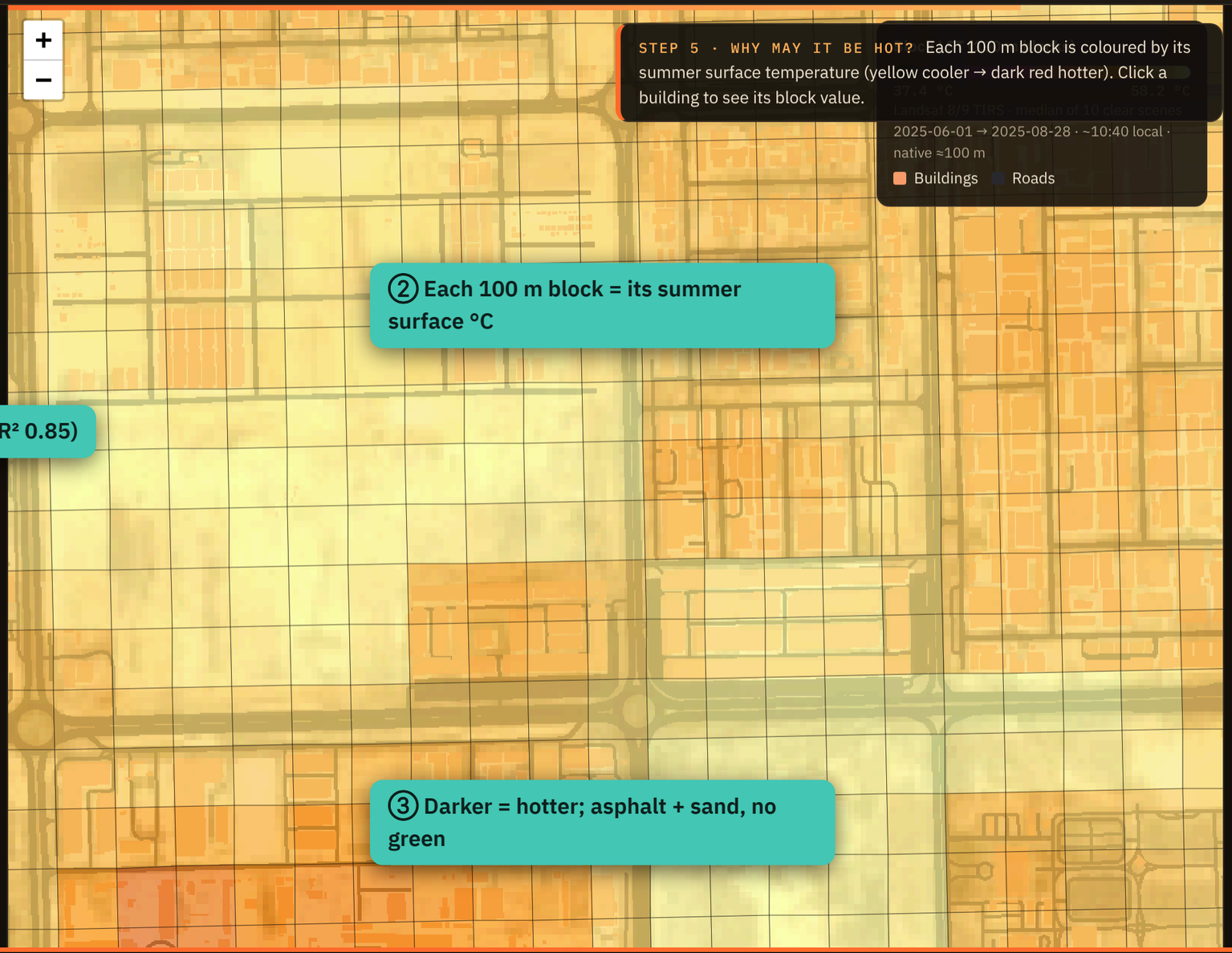
0.85 R², spatial CV

1.2 °C mean error

① Model links surfaces to heat (R² 0.85)

Where to act first in Musaffah

1	M-001 · industrial outdoor workers Shade structures	57.2°
2	M-002 · delivery riders Shade structures	57.6°
3	M-003 · industrial outdoor workers Shade structures	57.3°
4	M-004 · delivery riders Shade structures	56.8°
5	M-005 · delivery riders Shade structures	57.6°



Musaffah deep dive

Abu Dhabi · 9 × 9 km at 10 m

WHERE	Persistent summer hotspots at 300 m
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WHY	Roof, road, sand or missing greenery, read from hyperspectral spectra
WHO	Residents + people outdoors: mosque-goers, bus riders, delivery riders, workers, children
WHAT NEXT	One named action per hotspot, with estimated °C and confidence

Open the last chapter, “Where to act first”, and pick a hotspot.

Open data: Planet Tanager (© 2025 Planet Labs PBC), Landsat 8/9, ESA WorldCover, WorldPop 2025,

STEP 6 OF 6 · WHAT SHOULD BE DONE?

Where to act first in Musaffah

M-001 = Musaffah hotspot #1, ranked by heat × people × missing shade. Ranked 100 m cells with the action, the reason in plain words, and the relative potential of trees, cool pavement and shaded stops.

→ Pick a hotspot from the list to see its card.

M-001 High

top

① M-001 = Musaffah hotspot #1 (rank by risk)

1	Shade structures	
2	M-002 · delivery riders Shade structures	57.6°
3	M-003 · industrial outdoor workers Shade structures	57.3°
4	M-004 · delivery riders Shade structures	56.8°
5	M-005 · delivery riders Shade structures	57.6°
6	M-006 · delivery riders Shade structures	57.7°
7	M-007 · industrial outdoor workers	57.8°



② Do-now plan matched to materials

HIGH | HIGH CONFIDENCE | RISK 82

Musaffah hotspot M-001 · Al Hayal Street

HEAT

57.2° block LST (100 m) · hotter than 93% of Musaffah

WHAT IS THERE

Road 54% · Roof 2% · Sand 44% · Green 0%

WHO IS OUTSIDE

industrial outdoor workers | pedestrians | 24 residents

ROOF MATERIALS WITHIN 300 M (86 BUILDINGS)

Concrete / cement 61% · Metal sheet (bare / painted) 18% · White / cool coating 14%

DO NOW · MATERIAL-MATCHED PLAN

1. Shaded rest node + water station; enforce MoHRE midday break 12:30–

◎ 05

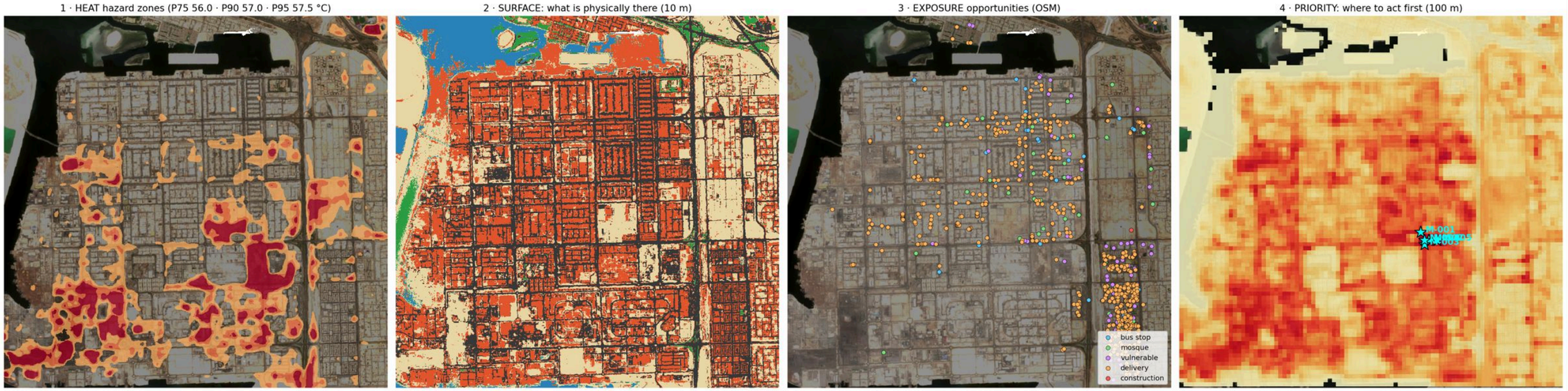
WHAT SHOULD BE DONE?

Hotspot M-001: one decision

Do first: shaded rest point + water, midday work break, trees and shade on the open sand (model -4.9 °C). Cool pavement is minor here (-0.1 °C).

EXAMPLE OUTPUT · NOTEBOOK FIGURE 12

Musaffah in four maps: heat → surface → people → priority



1 Heat

orange / red = elevated, high, extreme surface heat (Landsat, 10 summer scenes)

2 Surface · 3 People

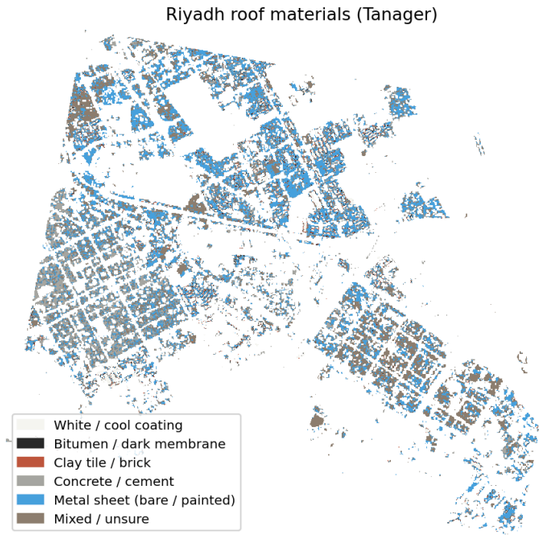
road, roof, sand, green at 10 m; bus stops, mosques, schools, workers

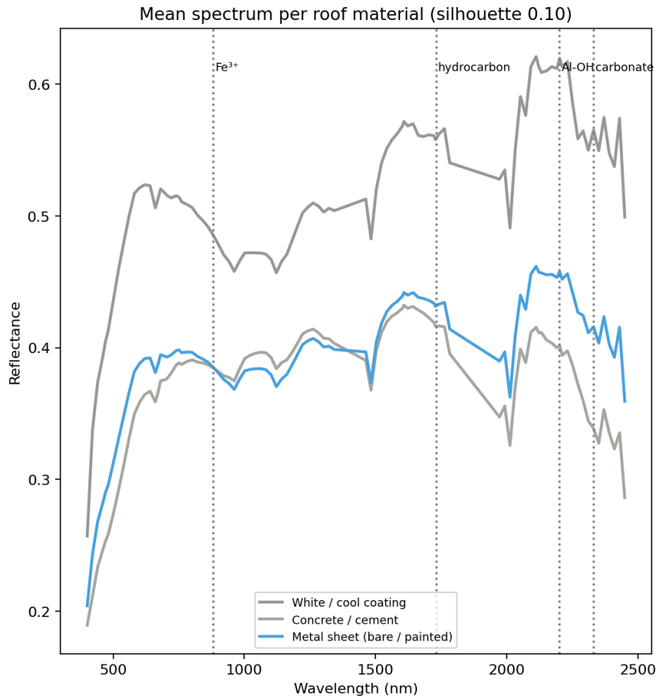
4 Priority

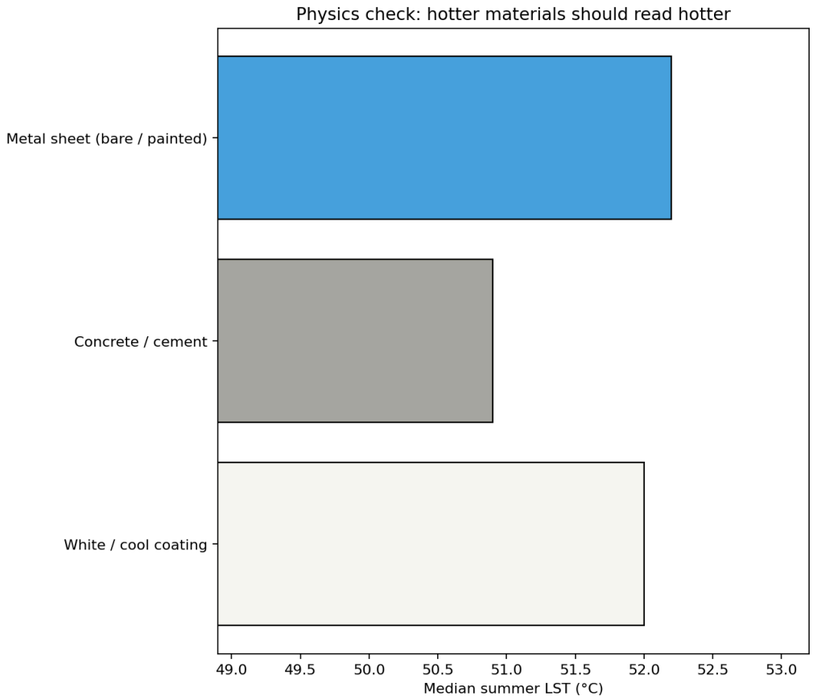
stars M-001…M-005 = where to act first

ROOFING MATERIALS FROM SPECTRA

Metal, concrete, tile, bitumen, white: the roof decides the fix







5

roof material classes from spectra

Rough

roof shares are rule-based estimates (0.26 agreement with the roof labels)

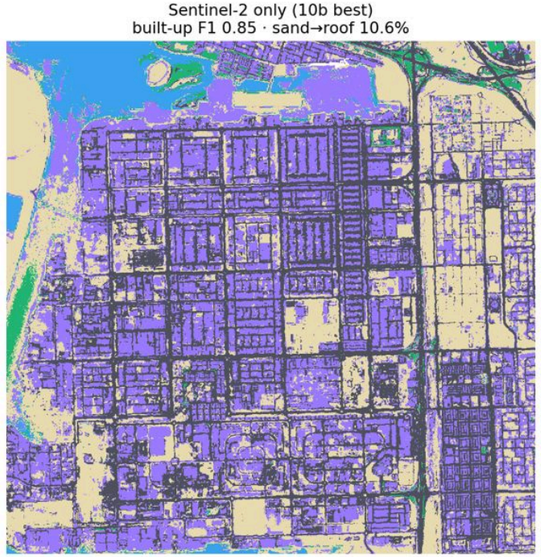
49

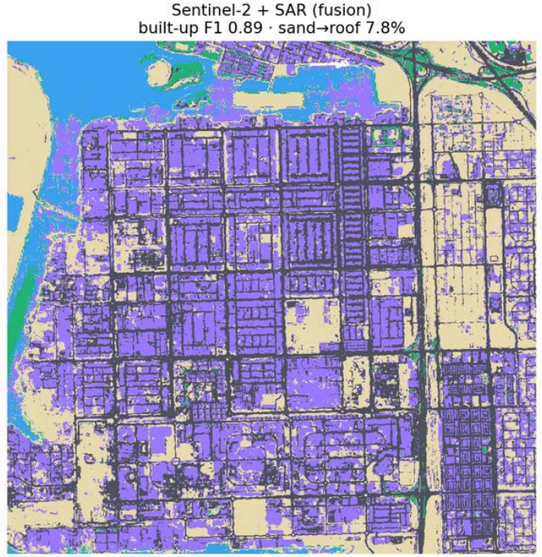
metal/bitumen roofs in Musaffah's high-heat zones

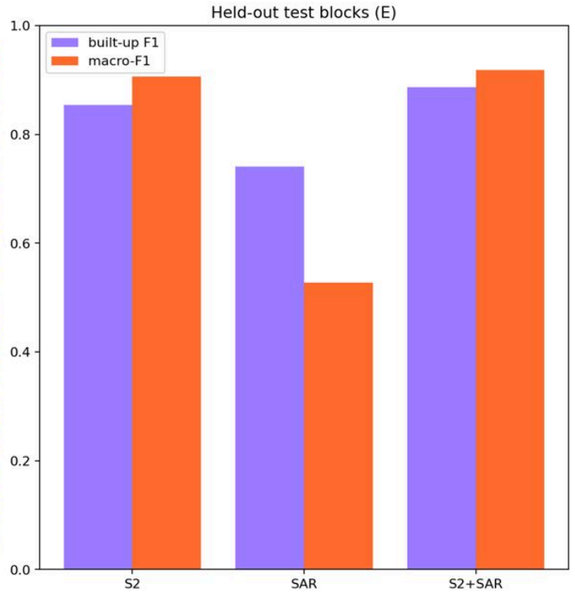
SENTINEL-1 RADAR FUSION

Radar sees structure: sand stops looking like a city









0.89
built-up F1 with SAR (S2 alone 0.85)

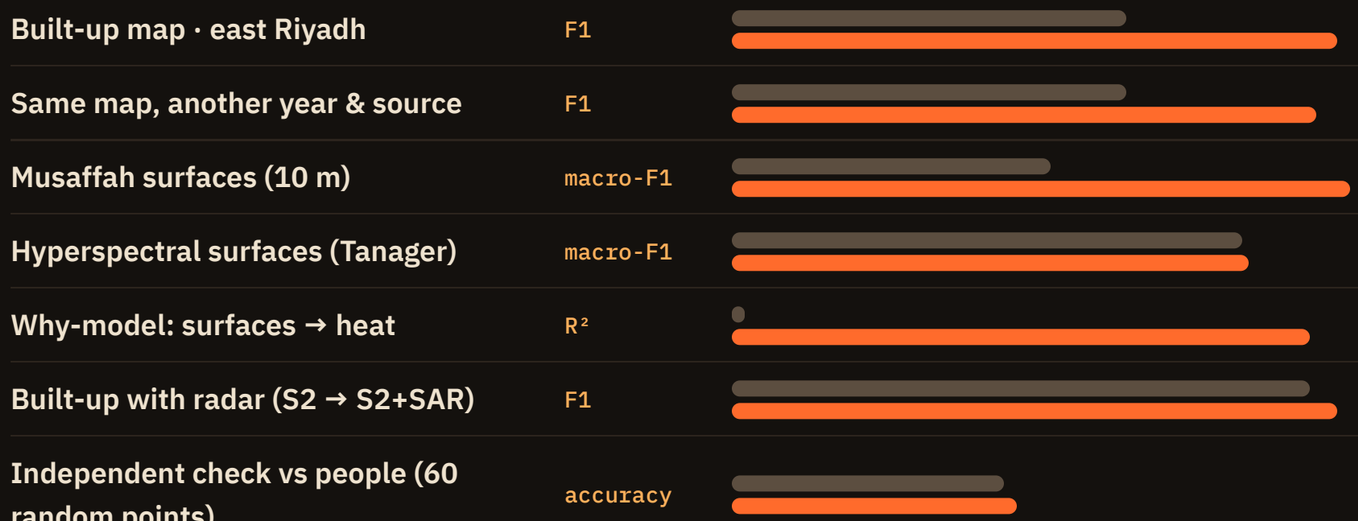
7.8%
sand mistaken for roof (was 10.6%)

+1.0 dB
radar brightening on buildings new since 2017

VALIDATION · AGAINST BASELINES AND HELD-OUT PLACES

Measured against baselines, on places the model never saw

starter / baseline · QAYDH



Built-up map · east Riyadh	F1	~~0.58~~ → 0.89	5-fold spatial-block CV vs ESA WorldCover
Same map, another year & source	F1	~~0.58~~ → 0.86	independent Impact Observatory 2023
Musaffah surfaces (10 m)	macro-F1	~~0.47~~ → 0.91	held-out 1 km test blocks
Hyperspectral surfaces (Tanager)	macro-F1	~~0.75~~ → 0.76	6 broad bands → full spectrum, unseen tiles
Why-model: surfaces → heat	R^2	~~0.00~~ → 0.85	spatial CV · RMSE 1.78 °C · MAE 1.19 °C
Built-up with radar (S2 → S2+SAR)	F1	~~0.85~~ → 0.89	held-out blocks · sand→roof 10.6% → 7.8%
Independent check vs people (60 random points)	accuracy	~~0.40~~ → 0.42	baseline = always 'Bare soil' · 95% CI 0.30–0.54 · κ people 0.85

MAE 1.4–1.5 °C
ERA5 vs 3 NOAA stations, air temp, hourly n≈2,000; used only to count danger hours

0.81
built-up IoU (Riyadh)

r 0.90
hotspots repeat 2024 ↔ 2025

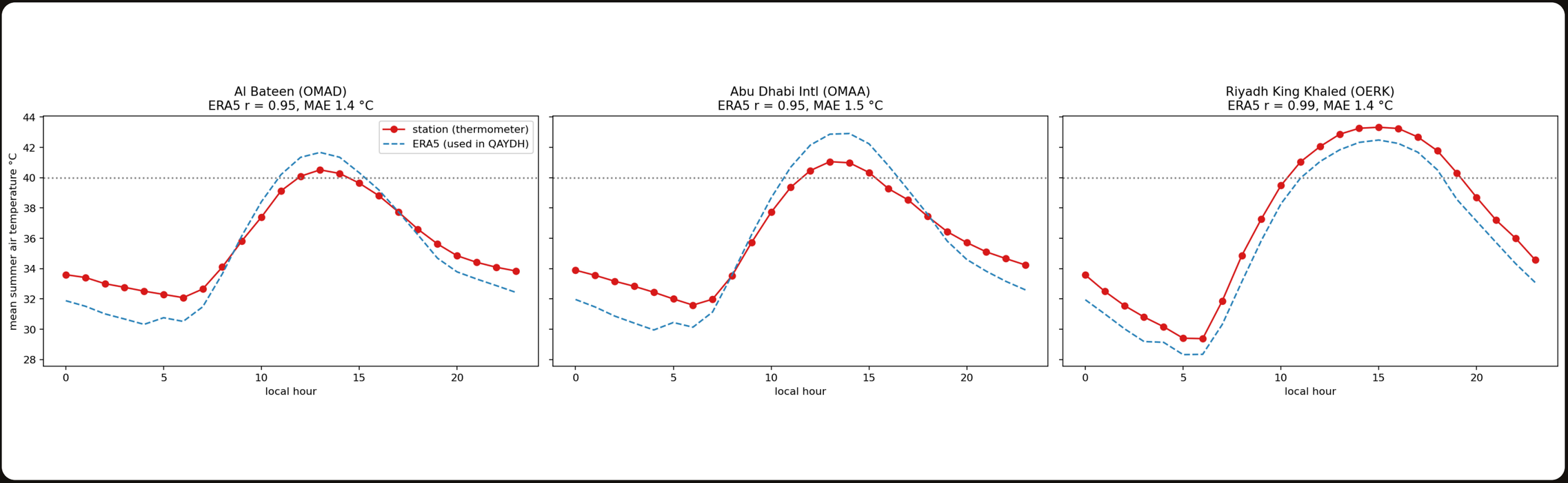
60–100%
top-20 stable under 4 weightings

×1.4
informal screen enrichment (p 0.023)

Musaffah 0.91 is macro-F1 on a near-balanced test sample (up to 3,000 pixels per class; majority-class baseline 0.21), not the district's natural class mix.

GROUND TRUTH · NOAA WEATHER STATIONS

Surface vs air, reported as a difference, not a validation



+15.2 °C

Al Bateen (OMAD): surface minus air on 18 Landsat days · ERA5 MAE 1.4 °C over 2,022 h

+15.5 °C

Abu Dhabi Intl (OMAA): surface minus air on 10 Landsat days · ERA5 MAE 1.5 °C over 2,019 h

+10.7 °C

Riyadh King Khaled (OERK): surface minus air on 22 Landsat days · ERA5 MAE 1.4 °C over 2,015 h

KNOWN TRADEOFFS

Being upfront

LABELS

Labels are cleaned, not assumed. 1,718 noisy labels removed automatically; 139 roof-material labels are AI-assisted and not yet human-checked. Blind check by 3 people on 60 random points: people agree (κ 0.85) but the 10 m surface map scores 0.42 at the exact pixel (0.65 within 10 m) vs a 0.40 baseline, so per-pixel classes are indicative; we decide on 100 m zones. No buffer between test blocks.

HEAT

Surface runs 11–15 °C above air (NOAA stations). We rank zones; ERA5, checked against stations, sets the hours.

MATERIALS

Tanager (30 m) separates materials clearly in Riyadh; EMIT (60 m) over Musaffah mixes roofs and roads in one pixel. Satellite 813 brings the finer detail.

SOLUTION VALUE & IMPACT

Value you can measure

157,076

residents in the hottest fifth of east Riyadh (transfer test)

445

cool-roof candidates in Musaffah

8

exposure sites at high or very high priority

−0.70 °C

per +0.10 roof albedo (95% CI)

Aligned with UAE Net Zero 2050 · We the UAE 2031 · Abu Dhabi urban heat goals · SDG 3 · 11 · 13

NEXT STEPS · INCUBATION

From proof of concept to a city service

0–3 MONTHS	Human-labelled training data (the blind check showed mixed 10 m pixels are the weak point) and human roof labels; field-check 20 Musaffah hotspots with the municipality.
3–6 MONTHS	Satellite 813 / MBZ-SAT hyperspectral over Abu Dhabi, Dubai and Al Ain; afternoon thermal from ECOSTRESS.
6–12 MONTHS	Municipal pilot in Musaffah: shade, cool roofs and rest points placed from QAYDH, then measured next summer.
PLATFORM	Could be hosted on Space42 gIQ as a dashboard + GeoJSON API + summer heat report.

CREDITS & ATTRIBUTION

Built on open data and open models

HYPERSPECTRAL	Tanager STAC Data, available at www.planet.com/data/stac, © 2025 Planet Labs PBC, All Rights Reserved. NASA EMIT courtesy of NASA LP DAAC.
SATELLITES	Contains modified Copernicus Sentinel data 2025 (Sentinel-1, Sentinel-2). Landsat 8/9 courtesy of USGS. ERA5 © ECMWF / Copernicus Climate Change Service.
MAPS & PEOPLE	© OpenStreetMap contributors (ODbL) · Microsoft Building Footprints (ODbL) · WorldPop · ESA WorldCover · Impact Observatory · NOAA ISD.
AI MODELS	Built with Llama (Llama-3.3-70B, Llama 3.3 Community License) · Segment Anything, Meta (Apache-2.0).
CODE	MIT licence · Team T0049: Entesar Alhebsi, Noora Alhajeri, Maryam Albonni.

QAYDH

From “where is it hot?” to “where we act first, why, and how.”

Thank you · Team T0049, United Arab Emirates

Live dashboard: noora-alhajeri.github.io/QAYDH-813-T0049 · Code: github.com/Noora-Alhajeri/QAYDH-813-T0049